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**DEVELOPMENT OF COOPERATIVE MANAGEMENT SYSTEM FOR
KINGSLAND PALA-PALA MULTI-PURPOSE COOPERATIVE
AND TRANSPORT SERVICES**

A CAPSTONE PROJECT

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**In Partial Fulfilment
of the Requirements for the Degree
Bachelor of Science in Information System**

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CHAPTER 1

INTRODUCTION

Cooperatives are important in financial help for the development of socio-economic development by providing accessible financial guide services including savings, lending programs, and investment opportunities. They operate on principles of mutual support and shared responsibility, enabling members to collectively benefit from financial resources and services that might otherwise be inaccessible (International Cooperative Alliance, 2024).

A Management System is one of the important roles in organizational operations by providing a centralized and structured approach in handling processes and information in order to support the operations of an organization. Integrating a digital management system with enterprise resource planning systems provides a solution for optimizing the workflow and transparency of an organization (Орлов 2024).

Broadly speaking, the implementation of the management system provides efficiency and transparency improving the quality of the organizational responsiveness in a dynamical environment.

Kingsland Pala-Pala Multi-Purpose Cooperative and Transport Services (KPMPCATS) is a multi-purpose cooperative that provides financial guide and



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transport-related services for its members. Their current operations include membership registration, lending programs, savings management, and the management of share capital contributions.

Currently, the cooperative is operating using a semi-manual system. This means that most of the processes in their current operation including registration for members, processing lending programs, deposit and withdrawal of savings, and deposit and withdrawal of share capital are handled using paper-based forms and manual documentation. Partnered with the cooperatives' accounting system as the only current digital tool. This system is used to manage financial recording and reporting.

In particular, web-based management systems enable organizations to provide real-time services, streamline financial record management, and improve communication between administrators and clients. The implementation of computerized loan and savings systems has been shown to enhance transaction accuracy, minimize processing delays, and improve member satisfaction (Laguda, 2023; Lestari & Magfiroh, 2025).

The researchers are motivated to accomplish the intended outcome of this research project to apply their knowledge in information systems to actual-world development, especially within the community. With the development of a



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Cooperative Management System, the researchers aim to contribute a meaningful solution to support organizational service quality.

Project Context

The client of this study is the Kingsland Pala-pala Multi-Purpose Cooperative and Transport Services (KPMP-CATS) located in Kingsland Village, Pala-Pala, Dasmariñas, Cavite. Founded in 1992, the cooperative generates income through transport services, capital investments, savings contributions and lending programs. Its membership consists of driver and allied workers, driver-operators, and investor associates. The cooperative's membership and transaction volume have grown over time, which has led to increased demands on the organization.

KPMP Cooperative currently uses eKoopbanker, an offline system developed by the National Confederation of Cooperatives (NATCCO), to keep track of its members, figure out loan interest, and make detailed reports. The eKoopbanker operates through a main computer unit which serves the server. When the server is turned off, the other connected computer units are unable to function, causing interruptions in daily operations. Only the administrative staff can access the system locally within the cooperative office. In addition, the KPMP also used a text-blast system to disseminate announcements. However, messages are sent to all members, requiring the cooperative to purchase mobile load, regardless of whether the information is relevant to specific members. And it is no longer



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functional, limiting the cooperative's ability to communicate updates to members efficiently.

Despite having an existing system, there are still underlying operational problems. Members do not have direct access to their financial records such as savings balances, share capital contributions, lending status, and transaction history. All inquiries must be processed through administrative staff, limiting transparency and increasing dependency on office visits or messaging. Furthermore, the cooperative maintains a large volume of physical documents to support transactions, creating storage limitations and difficulty in record retrieval as files pile up over time.

These inefficiencies result in long verification processes, increased administrative workload, and delayed communication. As membership continues to grow, the volume of inquiries, transaction processing, and record management also increases. This affects timely decision-making from management and members, this can also reduce the members' trust in the cooperative. The cooperative's current process flow system showed the inefficiencies that needed to be addressed to ensure better, transparent, accurate, reliable, and efficient operations.



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Purpose and Description

The proposed system is a web-based application that will serve as a platform that will allow the cooperative management and members to access services through a secure web interface that will improve the operational process. It will provide modules for member registration, profile management, savings and share capital monitoring, loan application processing, and announcement dissemination.

The proposed system will provide members with secure online access to their personal cooperative records. Through individual accounts, members will be able to view their share capital, savings balances, loan information, and transaction history. The system will also allow members to apply for lending programs and receive real-time announcements directly through the platform. This reduces the need for physical visits, minimizes repetitive inquiries, and improves communication efficiency.

The purpose of this is to provide a digital solution that will enhance operational efficiency, improve data accuracy, promote transparency, and strengthen communication within the cooperative and beneficiaries including;

Kingsland Pala-pala Multipurpose Cooperative, the primary beneficiary of this study will gain a system that will streamline operational processes



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Cooperative management and administrative staff will benefit from reduced manual workload, improve record accuracy, better data management, and accurate report generation.

Cooperative members will benefit from convenient access to their personal information, online services, and timely announcements without needing to visit the office.

Researchers will gain practical experience in system analysis, design, development, and evaluation, enhancing their technical and professional skills.

Future researchers will benefit from this study as a reference for developing similar web-based cooperative management systems, providing insights into system design, implementation, and evaluation best practices.



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Objectives of the Study

The general objective of the study is to design and develop a Web-Based Cooperative Management Information System that improves operational efficiency, enhances communication, and provides members with secure access to cooperative services and information.

Specifically, it aims to:

1. Design a Web-Based Cooperative Management Information System that is capable of providing the following general features to the members of the cooperative:

- a. A member registration and authentication module that allows members to securely create accounts and log in to the system;
- b. A member information module that enables members to manage their share capital, savings, lending records, and transaction history;
- c. A module that allows members to apply for lending programs, savings deposit and withdrawal request, and deposit and withdrawal request share-capital;
- d. An admin module that enables the managing department to view and manage the reports for share capital, savings, lending records, analytical records, transaction history, manage member accounts,



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review lending applications, assess eligibility, and approve or reject loan requests.

e. An announcement module that can be used by the management to provide real-time cooperative announcements to members.

f. A reporting module that generates financial and operational reports, including savings summaries, lending records, transaction histories, and member reports.

2. Develop the proposed system using appropriate hardware and software requirements, including necessary development tools, programming

3. Evaluate the system based on established software quality standards, specifically using the ISO/IEC 25010 evaluation criteria.

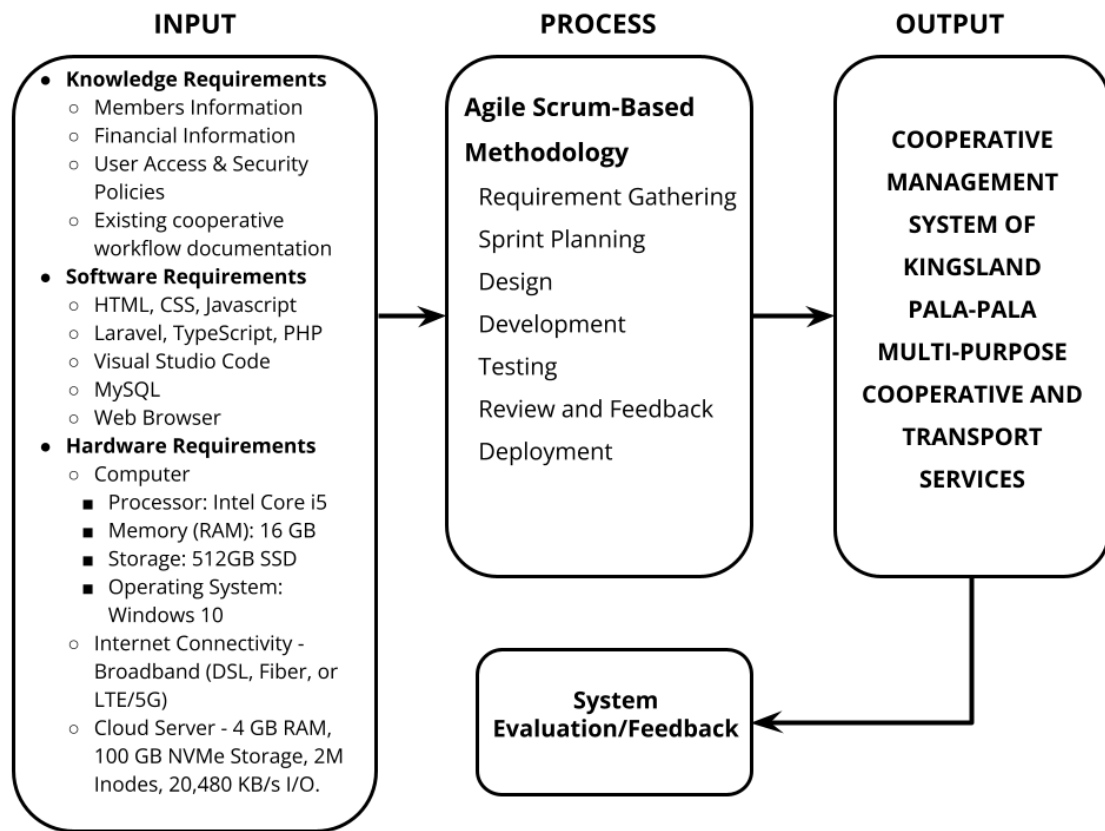
4. Prepare an implementation plan to guide the proper deployment, installation, and utilization of the system within the cooperative.



Conceptual Framework

Figure 1

Conceptual Framework



This study uses the Input-Process-Output (IPO) Model as the guiding framework for the development of the proposed Cooperative Management Information System for KPMP-CATS. The IPO model highlights the resources needed as inputs, the research problems, the methodologies and activities as processes, and the final system as the output are addressed.

The input section contains the requirements needed to develop the system. These include knowledge requirements such as members' information, financial



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information, user access and security policies, and existing cooperative workflow documentation. It also includes software requirements, which specify the technologies used to build the system, such as HTML, CSS, JavaScript, Laravel, TypeScript, PHP, Visual Studio Code, XAMPP (MySQL), and a web browser. Additionally, hardware requirements are listed, including a computer with an Intel Core i5 processor, 16 GB RAM, 512 GB SSD storage, Windows 10 operating system, and a stable internet connection like broadband or LTE/5G. Cloud server specifications are also included to support system deployment and performance..

The process will follow the Software Development Life Cycle (SDLC) using the Agile Scrum-Based model, the system will produce modular features which makes it the ideal for deploying a web-based system, this also emphasizes iterative and incremental development, frequent feedback, and adaptability to change throughout the project phases (Atlassian, n.d.).

The expected output is a fully functional web-based Cooperative Member System that provides efficient data management, reporting capabilities, and improved communication between cooperative management and members.



Scope and Limitations

This study focuses on the design and development of a Web-Based Cooperative Management Information System for Kingsland Pala-Pala Multi-Purpose Cooperative and Transport Services, operating in Dasmariñas City, Cavite. The system focuses on automating and centralizing the cooperative's core operations to improve accuracy, efficiency, and accessibility of records through a web-based platform.

The scope of the system includes the following functional areas:

- **User Roles and Access**

- Administrator
- Cooperative Staff
- Members/Drivers

Role-based access control is implemented to ensure secure system usage.

- **Member Registration and Authentication**

- Secure account creation and login for authorized users.

- **Member Information Management**

- Storing and managing member profiles, transaction history, and share capital records.

- **Savings and Share Capital Monitoring**

- Recording of member contributions with automatic balance updates.



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- **Lending Management**

- Lending application processing
- Automatic computation of loan interest and penalties
- Lending status tracking and payment recording

- **Reporting**

- Generation of savings reports, loan summaries, and transaction history reports.

- **Announcement Management**

- Posting of official announcements by authorized personnel for member access.

The system will be accessible through standard internet browsers and will be developed, tested, and implemented in accordance with the approved capstone project schedule.

The system is developed specifically for Kingsland Pala-Pala Multi-Purpose Cooperative and Transport Services and its members. The system does not include integration with external banking systems or E-wallet. Additionally, the system requires internet connectivity for access and operation.

The study is limited to a web-based application only. The development of a mobile application is not included, and access is restricted to browser-based platforms. The system focuses solely on basic cooperative operations, including membership registration and management, savings and share capital monitoring,



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loan processing, announcements, and reporting. Other functionalities such as inventory management, payroll processing, biometric attendance, and advanced accounting modules are excluded from the scope.

In terms of security, the system implements standard authentication and database security measures. However, advanced cybersecurity solutions such as penetration testing and enterprise-level encryption are not included in this study.



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Definition of Terms

Allied Worker. *(Operational)* A member who works within the cooperative, including drivers, dispatchers, and staff.

Cooperative/ Coop. *(General)* An organization owned and operated by a group of individuals for their mutual benefit.

Cooperative Management. *(General)* The people in charge of the management and office work inside the cooperative.

Investor Associate. *(Operational)* A member who invests in the cooperative but is not a driver or operator.

Lending Program / Loan. *(Operational)* A financial program through which members can borrow funds from the cooperative.

Member. A cooperative investor who may also serve as a Driver, Operator, or Allied Worker, in addition to holding the status of an Investor Associate.

Operator. *(Operational)* A member who owns and operates a transport vehicle under the cooperative.

Savings Contribution. *(Operational)* The amount a member deposits regularly into the cooperative savings.



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Share Capital. (*Operational*) The member's investment contribution to the cooperative.

Web-Based System. (*General*) A system that is accessed through an internet browser.



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CHAPTER 2

REVIEW OF RELATED LITERATURE AND STUDIES

Foreign Literature

A management system is an essential tool in order to improve the operation of an organization. These systems help growing organizations manage their records, monitor financial activities, and provide accurate information to the members and to the organization's management team. In cooperative organizations where many transactions happen regularly, a management system helps assure the data is properly organized and accessible.

Transparency in managing a cooperative is one of the most important aspects to maintain member trust in the cooperative organization. According to a study conducted by Ratnawati and Santoso (2024), transparency, especially in financial and operational management has a direct impact on increasing the level of trust within the cooperative members. Members that are able to look clearly in their financial records and transactions makes them become more confident in the management and operations of the cooperative. This shows that having a transparent system boosts relationships within its members maintaining a strong relationship between the management and its members.

Additionally, monitoring the performance and managing records effectively are important for the growth of the cooperative. Although many cooperatives



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demonstrate resilience, some organizations still fall back to a structured system for evaluating operational performance because of the limited technological support (Attolba-Aquino and Castañeda, 2025). This shows that a centralized system will be able to provide assistance to a growing cooperative that will also show transparency to its members.

The use of specialized management systems will further show that it improves the effectiveness of managing a cooperative. A study showed an 80% reduction in need for physical documents and physical storage after implementing a digital and cloud archive (Hafid & Wakhid, 2025).

Overall, the literature suggests that the management system is an essential tool to have for the cooperatives because this helps improve transparency, strengthen member trust, and support non physical storages.

A lengthy process is both inefficient for the cooperative management and for their new signing members, which is why automating the cooperatives' operations will improve their operational efficiency. Automating the process allows the organization to reduce the lengthy process they currently have. In a membership-based organization such as the cooperative, automation will help them manage large numbers of people signing up to the cooperative, record information, and record transactions more efficiently.

A study explained that the workflow of an automated system is designed to reduce operational bottlenecks and can improve service delivery. These systems



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allow organizations to automate managing tasks, which helps the management staff focus more on other important responsibilities (Kalluri 2024).

Another benefit from this is the ability to address the storage and documents management challenges. According to a study, the digital adoption of an organization involves replacing the paper-based process with a centralized digital one. The digital system will allow records to be stored electronically, which reduces the need for a physical storage space and simplifies document retrieval (Van Veldhoven and Vanthienen, 2023). Digitalizing these records helps solve these limitations by organizing data in a centralized and searchable automated system.

Local literature

A qualitative case study by Rabanes and Janamjam (2025) examined the role of information technology (IT) and the challenges faced by multipurpose cooperatives as a foundation for designing an IT system. The study involved six registered multipurpose cooperatives in Bukidnon, with nine participants from management, personnel, and staff taking part in interviews. The collected data were transcribed, coded, and analyzed thematically. Findings showed that IT integration in cooperatives primarily supports functions such as the use of Microsoft applications, communication processes, general accounting, data management, and operational improvement. However, several challenges were identified, including issues related to accessibility, system integration, and information security. These findings served as the basis for developing the



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Cooperative Upgrade Management Information System, which was designed to be user-friendly, cost-efficient, secure, and accessible even in areas with limited internet connectivity (Rabanés & Janamjam, 2025).

Similarly, The study of Hilbero and Bitonio (2025) examined the modernization efforts of transport cooperatives in Bacoor City, Cavite, focusing on fleet upgrading, consolidation, and compliance with regulatory requirements. Findings showed that modernization contributed to improvements in operational efficiency and safety; however, many cooperatives, especially smaller ones, experienced challenges such as financial limitations, complex loan processes, and limited technical capacity. The study highlighted the importance of stronger financial support, training programs, and coordinated planning to ensure effective and sustainable cooperative modernization (Hilbero & Bitonio, 2025).

Another study by Abellana et al. (2023) examined the organizational, social, economic, and financial management performance of cooperatives in Negros Occidental, Philippines, using a descriptive-quantitative research design. Data gathered from cooperative officers and members showed that cooperatives generally achieved satisfactory to excellent performance levels. The findings indicated that organizational effectiveness, social responsibility, and economic practices significantly influence financial outcomes. The study emphasized the importance of strengthening management strategies and operational systems to support sustainable cooperative growth and financial stability.



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The study by Catapang et al. (2025) conducted a comparative analysis of management practices and digitalization among selected cooperatives in Lipa City, Batangas, Philippines. The research examined key management functions such as planning, coordinating, commanding, and controlling, alongside the level of digital transformation among cooperatives. Using statistical methods including ANOVA and confirmatory factor analysis, the findings revealed variations in management practices based on asset size, while differences in digitalization were influenced by the cooperatives' years of operation and membership size. The study emphasized the importance of targeted training programs in digital technologies for cooperative leaders and local government units to address operational gaps and enhance overall cooperative performance (Catapang et al., 2025).

Moreover, Tarin (2024) focused on the operations management practices of multipurpose cooperatives in the Rinconada area, Philippines, particularly their compliance with the Cooperative Development Authority's performance standards. Using surveys, interviews, and observations, the study found that cooperative performance is influenced by factors such as asset size, services offered, membership, and years of operation. It also identified operational challenges and recommended the development of a standardized operations manual to improve management efficiency and organizational effectiveness.



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Foreign Studies

In a study by Sudaifah and Putri (2021), the researchers created a Cooperative Management Information System for the Unwaha Cooperative to assist the cooperative transactions. Using the CodeIgniter framework PHP and MySQL, the system successfully automated approvals, allowing cashiers and admins to manage transactions more easily than with paper-based methods. However the results suggested that the system needs further maintenance to fix errors and improve the overall functionality.

Similarly, Wanga (2025) showed the design and implementation of a Management Information System for SACCOS to solve the challenges of manual financial management. The developed system also functions like member registration, deposits, withdrawal loan and payment modules, built using PHP, Apache, MySQL. Although the results show efficiency and faster flow of information. It shows that it has database management issues and delays in reports. These studies are useful for the researchers as it highlights the importance of having user levels for coop management and members, which helps in organizing various daily operations of the cooperative. It provides insights in developing important modules.

In addition, Rilwan et al (2022) developed a web-based loan application management system for a Nigerian educational institution. The technologies used



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for the implementation included MySQL, PHP, HTML, CSS and JQuery. The results showed that the system helped in eliminating delays in administrative and errors from the manual encoding. It also made the process of verifying and approving loan applications much faster. Although the only limitation of the research is that it focuses on managing loan applications, which does not include the other process of the cooperative. This study is helpful for our research as it provides the framework for developing the membership registration and lending module that are also crucial for KPMP Cooperative.

Another study by Hambali et al. (2024) focused on the design and evaluation of a Web-Based Membership Management for clubs and associations to automate tasks, centralize data and provide a module for both members and administrators to member registration, manage profiles and track financial transactions. The system was developed using PHP and HTML. As a result it shows satisfaction in the system's usability and efficiency of managing registration and financial transaction reports. Moreover, it shows limitations on data security concerns when the system is handling sensitive member information and transactions. This study is important as it contributes to membership registration and management.

Moreover, Saputra et al (2025), designed a web-based cooperative information system for the Korong Gadang Cooperative in Indonesia to address the inefficient manual encoding and spreadsheet methods that led to delayed reportings and frequent errors. It also features modules for member management,



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savings and loan transaction processing, financial reporting and dashboard. It was also built using Laravel Framework and a MySQL database, that ensure secure and organized data handling. The results show improvement in operational performance, enhanced data accuracy and transparency.

However, this study also has several limitations, such as the system is restricted to desktop browsers, which limits accessibility for members who use mobile devices, it does not have notification features such as the email or SMS and insufficient cybersecurity measures. This study is highly relevant in developing the Cooperative Management System for KPMP Cooperative. It shows how a web-based platform can improve operational efficiency and transparency, as the members and staff will have access to their records in real-time.

Local Studies

Cooperative organizations rely on information technology to improve their operational performance. Rabanes and Janamjam conducted a study in 2025 which examined how multipurpose cooperatives use information technology to address their management needs and operational difficulties. The study found that the use of information technology improves various operational areas such as communication, general accounting, data management, and process improvement within cooperatives. Digital tools and computerized systems enable cooperatives to handle financial transactions while keeping precise records and providing better services to their members. Organizations use technology to boost their productivity



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while making their operations more efficient and enhancing their decision-making capabilities. The findings showed that organizations which implement information technology solutions can achieve better transparency and operational efficiency which results in improved overall performance for their organizations. increasing transaction processing accuracy. The researchers found that using computerized management systems offers major operational benefits to cooperatives and enhances service performance **(Rabanes & Janamjam, 2025)**.

Another study by Geroso and Maguate (2023) in Negros Occidental has examined the performance of organizational, social, economic, and financial of cooperative. The researchers gathered information from ninety-nine cooperatives through interviews and performance reports using a descriptive-quantitative method. The results indicated that cooperatives showed excellent organizational and economic performance and the outcome in social and financial terms achieved “very satisfactory”. The study found that effective management practices help organizations achieve financial stability and sustain operations. It is recommended that cooperative organizations establish ongoing monitoring systems and performance improvement programs to enhance operational performance **(Geroso & Maguate, 2023)**.

Similarly, one study by **Aligno et al. (2025)** showed that using traditional manual processes in cooperatives often lead to delays, inaccuracies, and difficulty in tracking financial records and adopting computerized financial management



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systems help improve operational efficiency. The researchers developed a computerized financial management system to manage all cooperative financial transactions which include loans and savings and other financial activities. The digital system implementation enabled better monitoring of member transactions which resulted in fewer financial calculations errors. The system also provided better financial recordkeeping because it enhanced accuracy and organizational capabilities. This development enabled cooperatives to deliver their services to members with increased speed and transparency. According to the research findings, the use of computerized management systems in cooperative organizations leads to better operational efficiency and higher member satisfaction.

Another related study from Nueva Ecija titled **Management Practices of Cooperatives in Nueva Ecija: A Basis for a Sustainable Development Program** examined how cooperatives implement management practices through their planning, organizing, directing, and controlling operations which executive leaders use to run their organization. Their study found that effective leadership and proper organizational structure are essential factors in maintaining efficient cooperative operations. By strengthening management strategies the service delivery and organizational performance can improve within the organization.

In addition, the study titled **Loans and Savings Management System for Bentuco National High School Faculty Cooperative** demonstrated how



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researchers developed a computerized system that automates all financial operations for a cooperative organization. The system included different modules for member registration, loan processing, savings monitoring, and report generation. This system provided benefits by improving financial record management, reducing paperwork, and increasing transaction processing accuracy. The researchers found that using computerized management systems offers major operational benefits to cooperatives and enhances service performance.

Related systems

This section presents existing systems related to the proposed Web-Based Cooperative Management System. The existing systems provide main functions which include member management and loan processing and financial monitoring that match the features of the proposed system. The system review process enables us to discover system strengths and weaknesses which serve as design guidelines for our proposed system.

The study Development of Digital Information Systems for Operational Efficiency of Savings and Loan Cooperatives by Saputra et al. (2025) demonstrates a system which enables the creation of joint information-sharing systems. The researchers created an online cooperative information system which helps savings and loan cooperatives that still use manual record-keeping systems to enhance



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their operational performance. The system consists of multiple components which include member management and savings and loan transactions and financial reporting and an administrative monitoring dashboard. The Rapid Application Development (RAD) method was used to create the system which enables users to develop systems through its iterative design process while users take part in the system development process. The study results established that the system brought about substantial operational efficiency enhancements because it decreased financial report preparation time from multiple days to one day and it improved data accuracy and transparency for cooperative members.

The research study Smart Financial Management for Cooperatives: A Web and Payment Gateway Integration Approach developed by Hakiki et al. (2025) presents another interconnected system. The system developed a web-based financial management system which included a payment gateway to enable cooperative members to access savings and loan services. The application developed by the team enables members to create accounts and manage their finances through online savings deposits and loan applications and loan payment processing which supports various payment methods including bank transfers and electronic payment systems. The system enables automated financial reporting and secure transaction processing which permits cooperative administrators and members to access real-time financial information. The study concluded that



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web-based cooperative systems which use online payment technologies will provide improved service accessibility and operational efficiency together with enhanced financial transparency.

Another system operates to digitize the process of managing cooperative records. The research study investigated how digital tools and computerized systems enhance recordkeeping procedures at cooperatives. The system enables cooperatives to handle their documents and financial records through digital platforms which improve their operational processes. The research results demonstrated that digital technologies accelerated information processing while enabling better data management and increasing operational transparency within cooperative organizations. The research findings demonstrate that automated systems need to be implemented by cooperative organizations to achieve their goals of operational efficiency and transparent operations (Halog et al., 2025).

Another related system, the eKoopbanker System, functions as a cooperative management software solution which the National Confederation of Cooperatives (NATCCO) developed to help cooperatives handle their financial and membership management tasks. The system provides support for essential cooperative activities which include managing member records and tracking savings and calculating loans and producing financial statements. The administrative staff uses the centralized database to record transactions and calculate loan interest and



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create essential financial reports which track cooperative financial operations. The cooperative office uses a main computer unit as its server to enable other computer units to connect with the system through a local network. This arrangement allows cooperative staff members to execute administrative functions while they create and manage records in the office space. The eKoopbanker system operates as a record-keeping and financial management tool which multiple Philippine cooperatives currently use to handle member information and financial operations. The system operates only within local areas because its main server computer needs to stay operational for system functionality according to the National Confederation of Cooperatives (NATCCO, n.d.). The related system shows how computerized management systems can enhance the efficiency of cooperative operations. The proposed Web-Based Cooperative Management Information System allows members to access their records and submit loan applications and receive announcements through a web-based platform while the eKoopbanker system restricts access to its local network. This improvement will create better transparency and accessibility and facilitate better communication between cooperative leaders and their members.

According to Ogar et al. (2026) the study entitled "Design and implementation of a cooperative management system" addressed the persistent challenges faced by cooperative societies, specifically the reliance on labor-intensive manual processes that result in transcription errors, delayed loan



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processing, and a lack of financial transparency for members. To resolve these issues, the researchers developed a Cooperative Management System (CMS) featuring core functional modules such as member registration, savings tracking, and automated loan disbursement. A key finding of their research was that transitioning from manual documentation to a digital system with role-based access control significantly accelerated the loan-approval cycle and reduced data-entry errors. By emphasizing the importance of real-time data updates and a responsive user interface to enhance member engagement, this study supports the objective of the KPMP-CATS project to replace server-dependent, offline limitations with a more accessible web-based solution that promotes organizational transparency.

Project Synthesis

The proposed web-based Cooperative Management System has many similarities from the previous cooperative studies and literature. Like any other systems reported in a study, it aims to reduce paper storage usage and also reduce the use of paper-based documents and forms that can show a real-time digital updates, services, transactions, and boost the member satisfaction (Lagunda, 2023 and Lestari & Magfiroh, 2025). The studies showed an emphasis on transparency, accessibility, increase of trust, and to reduce papers in order to address the lack of



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big storage facilities; this lines up with the system that is currently being developed. According to a study, there is an anticipated 80% cut in physical document storage (Hafid & Wakhid, 2025), this only proves that a fully centralized-electronic system is effective and efficient. Furthermore, a study explains how important the automation's role in toning down the workload for the management staff, a benefit also built into the proposed lending processing (Kalluri, 2024).

A local study highlights IT integration for cooperative processes, in which the MIS directly adopts (Rabanes & Janamjam 2025). Comparable web-based designs in the Philippines and Indonesia share the same modules, confirming the proposal's relevance (Sudaifah & Putri 2021, Wang 2025, and Saputra et al., 2025).

Overall, the key difference is within the scope, the proposed MIS excludes mobile-app access, payment-gateway integration and advanced cybersecurity measures that appear in later studies. Consequently, the proposal aligns closely with prevailing literature on core cooperative digitization while deliberately limiting peripheral features for an initial, focused deployment.



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CHAPTER 3

METHODOLOGY

Research Approach

The study used a **developmental research method**, which focuses on the design, development, and evaluation of an information system intended to address real-world organizational problems. This method is appropriate for the study as it involves the creation of a **Web-Based Cooperative Management System** for Kingsland Pala-Pala Multi-Purpose Cooperative and Transport Services (KPMP-CATS), aiming to improve operational efficiency, data management, and communication processes.

The research is guided by the **Input-Process-Output (IPO) Model**, which serves as the conceptual framework of the study. The IPO model structures the system development into three major components: inputs, processes, and outputs. The input includes system requirements such as member information, financial data, and security policies. The process involves the methodologies and development procedures applied, while the output represents the completed Cooperative Management System. This framework ensures a systematic flow from problem identification to system implementation.



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The study adopted the **Software Development Life Cycle (SDLC)** using an Agile Scrum-Based methodology. This approach was selected due to its ability to support iterative and incremental development, allowing continuous improvement of the system based on user feedback. The Agile Scrum model is particularly suitable for web-based systems, as it promotes flexibility, adaptability, and frequent evaluation throughout the development process

Additionally, the **development strategy** of the study follows an **iterative and incremental approach**, where the system is developed in phases through sprint cycles. This strategy allows continuous testing, evaluation, and integration of user feedback, ensuring that the system meets the needs of the cooperative and improves its overall functionality and usability.

Requirements Analysis

This outlines the functional requirements of the system. Information must be captured from the user perspective. This describes the details of the current system (who are the people involved, what are the business activities, where the environment takes place, when is the timing, and how procedures are performed.

The output involves Use Case diagrams, descriptions of each use case, interface designs, data flow diagrams and story board showing how the software would look like.



Current Flowchart **BIANCA**

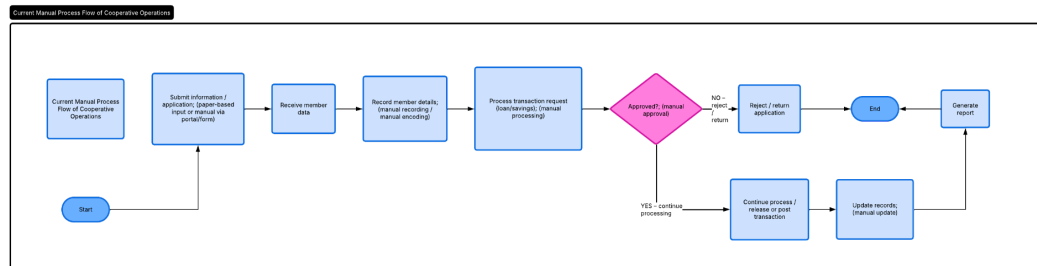


Figure ?. **Current Flowchart.** This presents the existing manual workflow of cooperative operations, in which members provide information and administrators handle the recording, processing, and approval of transactions manually. It includes a decision stage for approval and ends with record updating and report generation, emphasizing the limitations that the proposed system aims to improve.

Use-Case Diagram **BHEA**

Proposed Flowchart **BHEA**

Theoretical Framework **GUILA**

Context Diagram **GUILA**

Data Flow Diagram for each user

Member **GUILA**

Management Staff **BHEA**



Admin

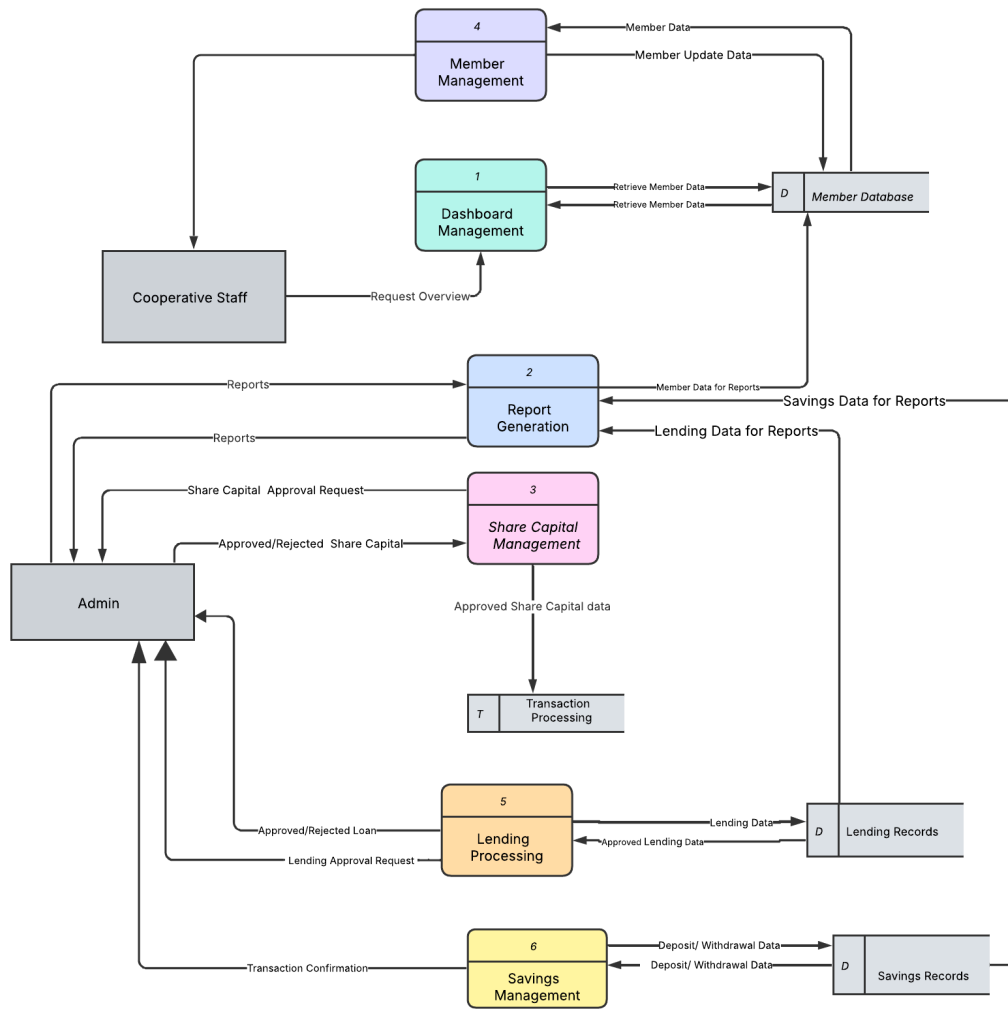


figure 3. **Data Flow Diagram (Level 1).** This shows data flows within the system among Cooperative Staff, the Admin, and key processes such as member, lending, savings, share capital, and report management. All transactions pass through appropriate processes, with admin approval required for lending and share capital before being stored and used for reporting.



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Systems Requirements

This section lists and discusses the hardware, software, peopleware and network specifications used during development and implementation of the software project.

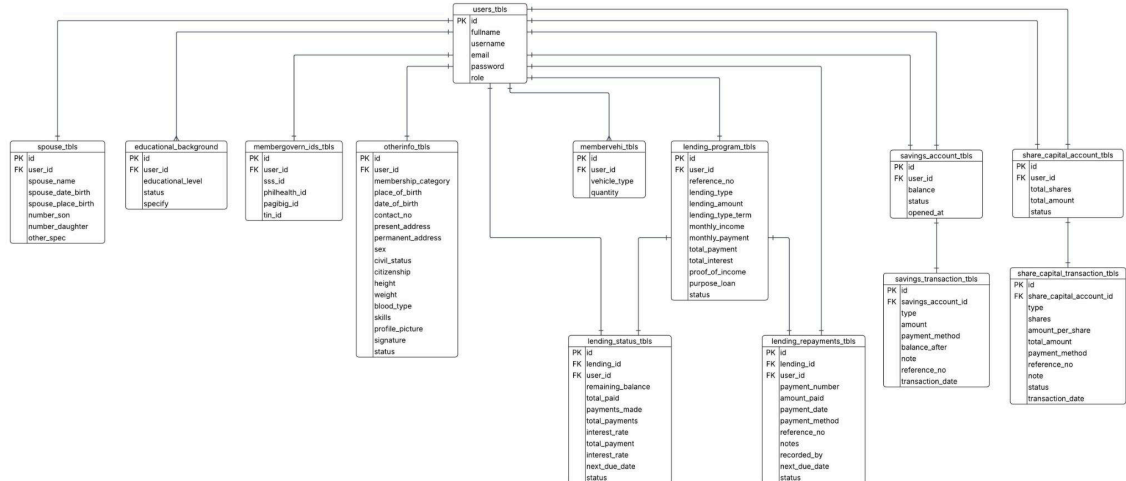
System Design

User Interface Design **Jhun**

Forms & Reports

Data Design **Rogelio**

Entity Relationship Diagram





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Data Dictionary

users_tbls

Field Name	Data type	Field length	Constraint	Description
id	int	20	PK, Auto Increment	Unique user id
fullname	varchar	255	NULL	User fullname
username	varchar	255	NULL	User username
email	varchar	255	NULL	User email
password	varchar	255	NULL	Encrypted password
role	varchar	255	NULL	User role
created_at	timestamp	-	AUTO	Record creation date
updated_at	timestamp	-	AUTO	Record update date

membervehi_tbls

Field Name	Data type	Field length	Constraint	Description
id	bigint	20	PK, Auto Increment	Unique member vehicle table id
user_id	bigint	20	FK	Foreign key user id
plate_no	varchar	255	NULL	User plate no
vehi_type	varchar	255	NULL	vehicle type



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quantity	int	11	NULL	how many vehicle
created_at	timestamp	-	AUTO	Record creation date
updated_at	timestamp	-	AUTO	Record update date

otherinfo_tbls

Field Name	Data type	Field length	Constraint	Description
id	bigint	20	PK, Auto Increment	Unique otherinfo table id
user_id	bigint	20	FK	Foreign key user id
membership_category	varchar	255	NULL	User membership category
place_of_birth	varchar	255	NULL	User place of birth
date_of_birth	varchar	255	NULL	User date of birth
contact_no	varchar	255	NULL	User contact no
present_address	varchar	255	NULL	User present address
permanent_address	varchar	255	NULL	User permanent address
sex	varchar	255	NULL	User sex
civil_status	varchar	255	NULL	User civil



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				status
citizenship	varchar	255	NULL	User citizenship
height	varchar	255	NULL	User height
weight	varchar	255	NULL	User weight
blood_type	varchar	255	NULL	User blood type
skills	varchar	255	NULL	User skills
profile_picture	long(text)	-	NULL	User profile picture
signature	long(text)	-	NULL	User signature
status	enum	Pending, Approved, Declined	Must be 'Pending Approved', 'Declined'	Indicates the current state of the request
created_at	timestamp	-	Auto	Record creation date
updated_at	timestamp	-	Auto	Record update date

spouse_tbls

Field Name	Data type	Field length	Constraint	Description
id	bigint	20	PK, Auto Increment	Unique family table id
user_id	bigint	20	FK	Foreign key user id
spouse_name	varchar	255	NULL	User spouse name



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spouse_date_birth	date	-	NULL	User spouse date of birth
spouse_place_birth	date	-	NULL	User spouse place of bith
number_son	int	11	NULL	User number of son
number_daughter	int	11	NULL	User number of daughter
other_spec	varchar	255	NULL	Stores another details
created_at	timestamp	-	Auto	Record creation date
updated_at	timestamp	-	Auto	Record update date

educational_tbls

Field Name	Data type	Field length	Constraint	Description
id	bigint	20	PK, Auto Increment	Unique educational tbl id
user_id	bigint	20	FK	Foreign key user id
educational_level	varchar	255	NULL	User educational level
status	varchar	255	NULL	current state of educational level



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specify	varchar	255	NULL	additional classification if user is not graduate
created_at	timestamp	-	Auto	Record creation date
updated_at	timestamp	-	Auto	Record update date

member_govern_ids

Field Name	Data type	Field length	Constraint	Description
id	bigint	20	PK, Auto Increment	Unique member govern ids id
user_id	bigint	20	FK	Foreign key user id
sss_id	longtext	-	NULL	User sss valid id
philhealth_id	longtext	-	NULL	User philhealth valid id
pagibig_id	longtext	-	NULL	User pagibig valid id
tin_id	longtext	-	NULL	User tin id
created_at	timestamp	-	Auto	Record creation date
updated_at	timestamp	-	Auto	Record update date

lending_program_tbls



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Field Name	Data type	Field length	Constraint	Description
id	bigint	20	PK, Auto Increment	Unique lending program id
user_id	bigint	20	FK	Foreign key user id
reference_no	varchar	255	NULL	User reference no
lending_type	varchar	255	NULL	User lending type
lending_amount	decimal	10, 2	NULL	User lending amount
lending_type_term	varchar	255	NULL	User lending type term
monthly_income	decimal	10, 2	NULL	User monthly income
monthly_payment	decimal	10, 2	NULL	User monthly payment
total_payment	decimal	10, 2	NULL	User total payment
total_interest	decimal	10, 2	NULL	User total interest
valid_id	longtext	-	NULL	User valid id
proof_of_income	longtext	-	NULL	User proof of income
purpose_loan	text	-	NULL	User purpose of loan
status	enum	Pending, Approved, Declined	Must be 'Pending Approved',	Indicates the current state of the status



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			'Declined'	
created_at	timestamp	-	Auto	Record creation date
updated_at	timestamp	-	Auto	Record update date

lending_status_tbls

Field Name	Data type	Field length	Constraint	Description
id	bigint	20	PK, Auto Increment	Unique lending status id
lending_id	bigint	20	FK	Foreign key lending id
user_id	bigint	20	FK	Foreign key user id
remaining_balance	decimal	10, 2	NOT NULL	User balance remaining
total_paid	decimal	10, 2	NOT NULL	User total paid
payments_made	int	11	NOT NULL	User payments made
total_payments	int	11	NOT NULL	User total payments
interest_rate	decimal	5, 2	NOT NULL	User interest rate
next_due_date	date	-	NOT NULL	User next due date
status	enum	Active, Completed, Overdue,	Must be 'Active, Overdue,	Indicates the current state of the status



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		Defaulted	Completed', 'Defaulted'	
created_at	timestamp	-	Auto	Record creation date
updated_at	timestamp	-	Auto	Record update date

lending_repayments_tbls

Field Name	Data type	Field length	Constraint	Description
id	bigint	20	PK, Auto Increment	Unique lending repaments id
lending_id	bigint	20	FK	Foreign key lending id
user_id	bigint	20	FK	Foreign key user id
payment_num ber	int	11	NOT NULL	User payment number
amount_paid	decimal	10, 2	NOT NULL	User amount paid
payment_date	date	-	NOT NULL	User payment date
payment_met hod	varchar	255	NOT NULL	User payment method
reference_no	varchar	255	NULL	User reference no
notes	text	-	NULL	User notes
recorded_by	bigint	20	NULL	who recorded is it by admin or user



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created_at	timestamp	-	Auto	Record creation date
updated_at	timestamp	-	Auto	Record update date

savings_account_tbls

Field Name	Data type	Field length	Constraint	Description
id	bigint	20	PK, Auto Increment	Unique savings account id
user_id	bigint	20	FK	Foreign key user id
total_shares	decimal	10, 2	NOT NULL	User total shares
total_amount	decimal	10, 2	NOT NULL	User total amount
status	enum	Active, Inactive, Closed	Must be 'Active, Inactive, Closed'	Indicates the current state of the status
created_at	timestamp	-	Auto	Record creation date
updated_at	timestamp	-	Auto	Record update date

savings_transaction_tbls

Field Name	Data type	Field length	Constraint	Description
id	bigint	20	PK, Auto Increment	Unique savings transaction id



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savings_account_id	bigint	20	FK	Foreign key savings account id
type	enum	deposit, withdrawal	Must be 'deposit, withdrawal'	Indicates the type savings transaction
amount	decimal	12, 2	NOT NULL	Stores the amount of money involved in the transaction
payment_method	varchar	255	NULL	Indicates the method used for the transaction
balance_after	decimal	12, 2	NOT NULL	Stores the remaining balance
note	varchar	255	NULL	additional remarks or comments
reference_no	varchar	255	NULL	Stores the reference no
transaction_date	date	-	NOT NULL	Stores the date when the transaction occurred
created_at	timestamp	-	Auto	Record creation date
updated_at	timestamp	-	Auto	Record update date

share_capital_account_tbls



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Field Name	Data type	Field length	Constraint	Description
id	bigint	20	PK, Auto Increment	Unique share capital account id
user_id	bigint	20	FK	Foreign key savings account id
total_shares	decimal	10, 2	NOT NULL	User total shares
total_amount	decimal	12, 2	NOT NULL	User total amount
status	enum	Active, Inactive, Closed	Must be 'Active, Inactive, Closed'	Indicates the current state of the status
created_at	timestamp	-	Auto	Record creation date
updated_at	timestamp	-	Auto	Record update date

share_capital_transaction_tbl

Field Name	Data type	Field length	Constraint	Description
id	bigint	20	PK, Auto Increment	Unique share capital transaction id
share_capital_account_id	bigint	20	FK	Foreign key share capital account id
type	enum	Subscription, Deposit, Withdrawal	Must be 'Subscription, Deposit, Withdrawal'	Indicates the type sharing their capital



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			Withdrawal"	transaction
shares	decimal	10, 2	NOT NULL	User shares in every transaction
amount_per_share	decimal	10, 2	NOT NULL	input amount per share
total_amount	decimal	12, 2	NOT NULL	total amount in overall shares in transaction
payment_method	varchar	255	NOT NULL	Indicates the method used for the transaction
reference_no	varchar	255	NOT NULL	Stores the reference no
note	text	-	NULL	additional remarks or comments
status	enum	Pending, Approved, Declined	Must be 'Pending, Approved, Declined"	Indicates the current state of the status
transaction_date	date	-	NOT NULL	Stores the date when the transaction occurred
created_at	timestamp	-	Auto	Record creation date
updated_at	timestamp	-	Auto	Record update date



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System Architecture

Network Model

Network Topology

Security

System Development

Software Specification

Hardware Specification

Program Specification

Programming Environment

Front End

Back End

Deployment Diagram



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Test Plan

System Testing Result

This describes the testing strategies that the system study will implement.

Unit Testing

Integration Testing

Compatibility Testing

Performance Testing

Stress Testing

Load Testing

System Testing

Acceptance Testing

System Evaluation Procedure

State what the criterion in ISO 25010 applicable in the capstone project evaluation. Insert table of matrix target evaluators who will evaluate the system.

Number of evaluators should be based on statistical result input in



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Chapter 4

RESULTS AND DISCUSSIONS

This section presents the discussion of findings that answer the research questions presented in Chapter 1. It also houses the qualitative or quantitative results that address the research problem and objectives.

Implementation Results

Describe the implementation plan and timeline. Describe the resources needed in the implementation of the solution.



CHAPTER 5

CONCLUSIONS AND RECOMMENDATIONS

Conclusions

This provides the general statements derived from the results of the capstone study. It also considers possible implications and limitations of the study and its findings.

Recommendations

These are the essence of conducting the study. These contain present recommendations based on the findings. It contains recommendations to the study sponsor and the actions to recommend based upon the data. It also highlights the recommendations to other researchers.



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REFERENCES

This section lists the scholarly and credible sources of information arranged in alphabetical order. The presentation of reference in the KLD research manuscript shall follow the American Psychological Association (APA) 7TH Edition Citation Guide.



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APPENDICES

This section provides additional information and documentation necessary for getting the whole picture of the conduct of the study. It also contains the materials used, correspondence, pictures, raw data, and other important attachments.

- A Correspondences (Letter of Request/ Endorsement)**
- B Data Collection Tool/s**
- C Informed Consent**
- D Signed Data Privacy Form**
- E Language Editor's Certificate**
- F. Statistician's Certificate (if applicable)**
- G. Documentation (Photos taken during the conduct of the study)**
- H. Accomplished Instrument Validation Form**
- I. Transcriptions/Computation (Samples)**
- J. Certificate of Originality**
- K. Ethics Review Certificate**
- L. Short Biographical Sketch of the Researchers**



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M. User Manual

N. Source Code of the Project

CAPSTONE PROJECT GRAMMARIAN CERTIFICATE

This is to certify that the capstone entitled,

(CAPSTONE PROJECT TITLE – in caps lock and inverted triangle)

authored by

(Authors name in alphabetical order – Last name, First name M.I.)



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has undergone editing and proofreading by the undersigned.

This Certification is being issued upon the request of (authors name) for
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(Signature above Grammarian Name)

Grammarian

Kolehiyo ng Lungsod ng Dasmariñas

(Date)

THESIS GRAMMARIAN CERTIFICATE

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(Authors name in alphabetical order – Last name, First name M.I.)

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Kolehiyo ng Lungsod ng Dasmariñas

(Date)

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(THESIS/CAPSTONE TITLE – in caps lock, bold and inverted triangle)



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authored by

(Authors name in alphabetical order – Last name, First name M.I.)

(Institute Name)

has been subjected to a similarity check on (Date) using Turnitin with a
generated similarity index of %.

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